

STATEMENT OF WORK/REQUIREMENTS DOCUMENT

Title: Air Force automated equipment for fiber power meter calibration

Requesting Lab: 686.01

I. BACKGROUND INFORMATION

We are purchasing a system of switches, shutters and attenuators from which to build an automated version for the optical power meter calibration setup. This automation equipment will be delivered to the Air Force as part of the Optical Fiber Modernization effort.

Purpose/Objective: This equipment will replace the > 20-year-old equipment in current use in Air Force optical fiber meter calibration service, as part of a larger modernization effort.

II. SCOPE

The Contractor shall deliver a quantity one system of equipment for fiber power calibration, inclusive of installation, warranty and training, and option line items.

III. MINIMUM REQUIREMENTS

The Contractor shall provide a system that meets all technical specifications identified below. All items must be new. Used or remanufactured equipment will not be considered for award. Experimental, prototype, or custom items will not be considered. The use of “gray market” components are not authorized for sale in the U.S. by the Contractor is not acceptable. All line items shall be shipped in the original manufacturer’s packaging and include all original documentation and software, when applicable.

Line Item 0001:

Description: Optical Fiber Automation Equipment

Quantity: 1

A. Technical Specifications for entire system

1. A compact system such as a mainframe with the following subcomponents
 - a. 3 x 1 fiber switch
 - b. 2 x 1 fiber switch
 - c. 2 fiber optical shutters
 - d. 2 optical attenuators.

The technical specifications for each of the subcomponents will be described separately.

2. The system must be optical fiber based, with SMF-28 fiber and FC/APC connections.
3. The system should be addressed by a single computer connection via GPIB
4. It must be possible to control the instrument via custom LabVIEW or Python software.
5. The sum of insertion losses of all subcomponents must be less than 8 dB. This can be achieved with for example < 1.4 dB insertion loss for every subcomponent, or by having some subcomponents with higher and some with lower insertion losses, summing to less than 8 dB.
6. All subcomponents must tolerate 50 mW input power.

B. Technical specifications for subcomponent A, 3x1 fiber switch

1. FC/APC input and output for SMF-28 fiber
2. The switch must connect any of at least three inputs to a single output. Larger numbers of inputs or outputs are acceptable, however.
3. The switch must operate over a range from 1260-1650 nm.

4. Must be computer controllable via GPIB.

C. Technical specifications for subcomponent B, 2x1 fiber switch

1. FC/APC input and output for SMF-28 fiber
2. The switch must connect any of at least two inputs to a single output. Larger numbers of inputs or outputs are acceptable, however.
3. The switch must operate over a range from 1260-1650 nm.
4. The switch must be repeatable to better than 0.1% over a period of an hour.
5. Must be computer controllable via GPIB.

D. Technical specifications for subcomponent C, optical shutter

1. FC/APC input and output for SMF-28 fiber
2. The shutter must block the light with better than 80 dB rejection in the wavelength range from 1260 nm – 1650 nm.
3. Tolerate 10 mW input power.
4. Must be computer controllable via GPIB.

E. Technical specifications for subcomponent D, optical attenuator

5. FC/APC input and output for SMF-28 fiber
6. Operate at 1310 and 1550 nm with attenuations from 0 to 75 dB possible.
7. Must be computer controllable via GPIB.
8. The total attenuation could be achieved (for example) with two attenuator modules in series if necessary.

IV. DELIVERABLES

<i>Description</i>	<i>Quantity or Format</i>	<i>Due Date</i>
Automation equipment	1	3-month lead time

Standards of Acceptance: The NIST TPOC or COR shall review (list deliverable) and respond with an acceptance or request for revision email to the Contractor Point of Contact (POC) within 10 days of receipt of deliverable. This section applies to deliverables which must be reviewed and approved for acceptances (draft plans, drawings, etc.)

V. PLACE OF PERFORMANCE

Air Force Primary Standards Laboratory (AFPSL), Ohio

VI. PERIOD OF PERFORMANCE/LEAD TIME

3 months after acceptance of offer

VII. DELIVERY TERMS

Delivery shall be F.O.B Destination and shall occur in accordance with the delivery due dates provided in the above table.

FOB Destination means: The contractor shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery. The contractor shall deliver all Line Items to:

National Institute of Standards and Technology
Shipping and Receiving
325 Broadway,
Boulder CO 80305

VIII. INSPECTION & ACCEPTANCE

In addition to the inspection and acceptance terms articulated in 52.212-4, the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

Performance Tests:

1. Switching ratio stability meets requirements.
2. Attenuation level meets requirements at both wavelengths.
3. Instrument can be controlled by custom LabVIEW VI via GPIB connection.

NIST may choose at its discretion to forego this part of acceptance testing.

A visual inspection of the equipment will be performed by the NIST TPOC to identify surface defects or any form of indication that any equipment was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the equipment due to defects and/or nonconformance. The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the equipment.

Ownership of the equipment shall transfer to NIST upon acceptance by the Government.

The Government will test, inspect, and accept or reject the equipment within (insert working days it will take to inspect, accept or reject) of the receipt of the equipment unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

IX. WARRANTY

The contractor shall warranty the entire system for a period of a minimum of 1 year after receipt of the equipment and shall be in accordance with terms in FAR 52.212-4. Warranty shall commence upon acceptance of the system by the Government.

X. PAYMENT SCHEDULE

Advance payment is not authorized. The Contractor must invoice in arrears according to the payment schedule. The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. 100% after installation and acceptance by the TPOC of fully installed system, AND
2. After the successful completion of the testing requirements set forth in this document under section set forth in this document, AND
3. After successful demonstration by the instrumentation that it performs IAW the technical requirements set forth in this document AND
4. After receiving an invoice submitted properly, IAW the purchase order terms and conditions.

NOTE: Partial shipments and partial invoices will not be accepted, unless otherwise requested and accepted by the Contracting Officer prior to award offer. Proposed payment schedules shall be submitted with vendor's response to the RFQ for consideration.